

114 學年度第二學期 離散數學 HW09

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1 題目

1.1 作業繳交說明

- 從 overleaf 下載 hw09.zip
- 從 overleaf 下載 hw09.pdf
- 建立一個新檔案夾，命名如下
- 離散數學_114590xxx_姓名_HW09
- 放入 hw09.zip
- 放入 hw09.pdf
- 將此檔案夾，壓縮成 zip 檔，檔名如下
- 離散數學_114590xxx_姓名_HW09.zip
- 將此壓縮檔，上傳到北科 i 學園 PLUS
- Do not share your homework with any living creatures.
- 作業遲交，一天內成績打五折，超過一天以零分計。

1.2 題號

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2 作答

2.1 page 608, chapter 9.1 Exercise 6

Determine whether the relation R on the set of all real numbers is reflexive, symmetric, antisymmetric, and/or transitive, where $(x, y) \in R$ if and only if

- (a) $x + y = 0$
- (b) $x = \pm y$
- (c) $x - y$ is a rational number.
- (d) $x = 2y$
- (e) $xy \geq 0$
- (f) $xy = 0$
- (g) $x = 1$
- (h) $x = 1$ or $y = 1$

- (a) • not reflexive. Since $1 + 1 \neq 0$.
- symmetric. Since $x + y = y + x$, it follows that $x + y = 0$ if and only if $y + x = 0$.
- not antisymmetric. Since $(1, -1)$ and $(-1, 1)$ are both in R .
- not transitive. For example, $(1, -1) \in R$ and $(-1, 1) \in R$, but $(1, 1) \notin R$.
- (b) • reflexive. Since $x = x$ or $x = -x$ is true for all x .
- symmetric. Since $x = \pm y$ implies $y = \pm x$.

- not antisymmetric. Since $(1, -1)$ and $(-1, 1)$ are both in R , but $1 \neq -1$.
- transitive. If $x = \pm y$ and $y = \pm z$, then $x = \pm z$.
- (c) • reflexive. Since $x - x = 0$ is a rational number for all x .
- symmetric. If $x - y$ is rational, then $y - x = -(x - y)$ is also rational.
- not antisymmetric. For example, $(1, 2)$ and $(2, 1)$ are both in R , but $1 \neq 2$.
- transitive. If $x - y$ and $y - z$ are rational, then $x - z = (x - y) + (y - z)$ is rational.
- (d) • not reflexive. Since $1 \neq 2(1)$.
- not symmetric. For example, $(2, 1) \in R$ but $(1, 2) \notin R$.
- antisymmetric. If $x = 2y$ and $y = 2x$, then $x = 4x \implies x = 0$, which means $y = 0$, so $x = y$.
- not transitive. For example, $(4, 2) \in R$ and $(2, 1) \in R$, but $(4, 1) \notin R$.
- (e) • reflexive. Since $x^2 \geq 0$ for all real numbers x .
- symmetric. Since $xy = yx$, it follows that $xy \geq 0$ if and only if $yx \geq 0$.
- not antisymmetric. For example, $(1, 2) \in R$ and $(2, 1) \in R$, but $1 \neq 2$.
- not transitive. For example, $(1, 0) \in R$ and $(0, -1) \in R$, but $(1, -1) \notin R$.
- (f) • not reflexive. Since $1 \cdot 1 = 1 \neq 0$.
- symmetric. Since $xy = yx$, it follows that $xy = 0$ if and only if $yx = 0$.
- not antisymmetric. For example, $(1, 0) \in R$ and $(0, 1) \in R$, but $1 \neq 0$.
- not transitive. For example, $(1, 0) \in R$ and $(0, 1) \in R$, but $(1, 1) \notin R$.
- (g) • not reflexive. Since $2 \neq 1$, so $(2, 2) \notin R$.
- not symmetric. Since $(1, 2) \in R$ but $(2, 1) \notin R$.
- antisymmetric. If $(x, y) \in R$ and $(y, x) \in R$, then $x = 1$ and $y = 1$, so $x = y$.
- transitive. If $(x, y) \in R$ and $(y, z) \in R$, then $x = 1$, so $(x, z) \in R$.
- (h) • not reflexive. Since $2 \neq 1$, so $(2, 2) \notin R$.
- symmetric. Since the condition is symmetric with respect to x and y .
- not antisymmetric. For example, $(1, 2) \in R$ and $(2, 1) \in R$, but $1 \neq 2$.
- not transitive. For example, $(2, 1) \in R$ and $(1, 3) \in R$, but $(2, 3) \notin R$.

2.2 page 619, chapter 9.2 Exercise 2

Which 4-tuples are in the relation $\{(a, b, c, d) \mid a, b, c, \text{ and } d \text{ are positive integers with } abcd = 6\}$?

$(6, 1, 1, 1), (1, 6, 1, 1), (1, 1, 6, 1), (1, 1, 1, 6), (2, 3, 1, 1), (2, 1, 3, 1), (2, 1, 1, 3), (3, 2, 1, 1), (3, 1, 2, 1), (3, 1, 1, 2), (1, 2, 3, 1), (1, 2, 1, 3), (1, 3, 2, 1), (1, 3, 1, 2), (1, 1, 2, 3), (1, 1, 3, 2)$

2.3 page 627, chapter 9.3 Exercise 14

Let R_1 and R_2 be relations on a set A represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \text{ and } \mathbf{M}_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Find the matrices that represent

- (a) $R_1 \cup R_2$
- (b) $R_1 \cap R_2$
- (c) $R_2 \circ R_1$
- (d) $R_1 \circ R_1$
- (e) $R_1 \oplus R_2$

- (a) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
- (b) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$
- (c) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
- (d) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
- (e) $\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

2.4 page 638, chapter 9.4 Exercise 20

Let R be the relation that contains the pair (a, b) if a and b are cities such that there is a direct nonstop airline flight from a to b . When is (a, b) in

- (a) R^2 ?
- (b) R^3 ?
- (c) R^* ?

- (a) The pair (a, b) in R^2 precisely when there is a city c such that there is a direct flight from a to c and a direct flight from c to b – in other words, when it is possible to fly from a to b with a scheduled stop (and possibly a plane change) in some intermediate city.
- (b) The pair (a, b) is in R^3 precisely when there exist cities c and d such that there are direct flights from a to c , from c to d , and from d to b – in other words, when it is possible to fly from a to b with exactly two intermediate stops.
- (c) The pair (a, b) is in R^* precisely when there is a sequence of flights leading from a to b – in other words, when it is possible to fly from a to b using any number of intermediate stops.

2.5 page 647, chapter 9.5 Exercise 24

Determine whether the relations represented by these zero–one matrices are equivalence relations.

- (a) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$
- (b) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$
- (c) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- (a) **No**, because it is not symmetric.
- (b) **Yes**, because it is reflexive (all diagonal entries are 1), symmetric (the matrix is symmetric with respect to the main diagonal), and transitive.
- (c) **Yes**, because it is reflexive (all diagonal entries are 1), symmetric (the matrix is symmetric with respect to the main diagonal), and transitive.

2.6 page 662, chapter 9.6 Exercise 8

Determine whether the relations represented by these zero–one matrices are partial orders.

- (a) $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$
- (c) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

- (a) **No**, because this relation is $(1, 1), (1, 3), (2, 1), (2, 2), (3, 3)$. It is clearly reflexive and antisymmetric. The only one pairs that might present problems with transitivity are the non-diagonal pairs, $(2, 1)$ and $(1, 3)$. If the relation were to be transitive, then we would also need the pair $(2, 3)$ in the relation.
- (b) **Yes**. Reasoning as it is reflexive (all diagonal entries are 1), antisymmetric (no two symmetric entries off the diagonal are both 1), and transitive (the only non-trivial composition path is $(3, 1)$ and $(1, 1)$, which yields $(3, 1)$, which is in the relation).
- (c) **No**, because it is not transitive; for instance, $(1, 3)$ and $(3, 4)$ are in the relation (represented by 1s in the matrix), but $(1, 4)$ is not (represented by 0).